

David Fernandez

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PROFESSIONAL SUMMARY

CSULB '25 Computer Science graduate experienced in Python/Node.js microservices and hybrid ML-powered event recommendations, seeking to design reliable, innovative software solutions.

PROJECTS

EventMate – Backend for Cross-Platform Mobile Application

Jan 2025 – Present

Flutter, Dart, Python, FastAPI, Sentence-BERT, MongoDB, Docker, AWS

- Conceptualized EventMate mobile app features, designed high-fidelity UI/UX prototypes in Figma, and coordinated with frontend developers on Flutter implementation.
- Designed and implemented Python/FastAPI microservices for hybrid recommendations, increasing event click-through by 15%.
- Handled data ingestion across 1,803 CA zip codes with haversine geospatial filtering, reducing API latency by 40%.
- Deployed JWT-secured services via Docker CI/CD, maintaining over 90% production availability.

ResNet-based Chest X-Ray Classification

Dec 2024

Python, PyTorch, NumPy, Matplotlib

- Fine-tuned ResNet50 CNN for Pneumonia vs. healthy classification, achieving 89.6% test accuracy and 0.94 ROC-AUC.
- Processed 5,000+ X-ray images with normalization and augmentation to improve model robustness.
- Documented preprocessing and evaluation in Jupyter Notebook for reproducibility.

End-to-End IoT Sensor Monitoring System

May 2024

Python, MongoDB, CLI

- Simulated three virtual IoT devices, with one to four sensor types (moisture meter, ammeter, thermistor, water consumption sensor), generating time-series data with per-sensor metadata.
- Built Python TCP server handling metadata filtering, data extraction, and calculations (average moisture, water usage per cycle, peak electricity in kWh).
- Implemented Python TCP client with menu-driven CLI, robust socket handling, and input validation for reliable querying.
- Deployed both server and client components on Google Cloud Platform VMs, ensuring high availability and simplified management.

PROFESSIONAL EXPERIENCE

Technic Modular Construction

Perris, CA

Modular Construction Specialist

Jul 2023 – Present

- Coordinated cross-functional teams to deliver modular building projects valued at \$100K+, consistently meeting deadlines and budgets.
- Ensured structural integrity and compliance with safety regulations, achieving zero incidents over 2 years.
- Maintained machinery, reducing downtime by 30% through proactive maintenance.

EDUCATION

California State University, Long Beach

Long Beach, CA

Bachelor of Science in Computer Science

May 2025, GPA: 3.5

- Relevant Coursework: Applied Machine Learning, Artificial Intelligence, Data Visualization.
- Awards: President's Honor List (Fall 2023; Fall 2024; Spring 2025).

TECHNICAL SKILLS

Languages: Python, C++, SQL

Data Stores: MongoDB, PostgreSQL

Spoken: English (native), Spanish

Frameworks: FastAPI, Sentence-BERT

Infrastructure: Docker, AWS (EC2), Git, Linux